

Formula One Championship Analysis Using SQL

1. Introduction

This project is based on the analysis of historical Formula One championship data using SQL. The system stores and analyses information related to drivers, constructors, engines, seasons, and championship performance from 1950 to 2024.

The project demonstrates how relational databases can be used for sports analytics and performance analysis. Using SQL queries, meaningful insights are generated such as championship dominance, constructor performance, driver statistics, and historical trends.

2. Objectives

The main objectives of this project are:

- To design a relational database for Formula One championship data.
- To perform data analysis using SQL queries.
- To identify top-performing drivers, constructors, and engines.
- To calculate performance metrics such as win percentage and dominance score.
- To understand the use of joins, aggregation functions, and relational database concepts.

3. Technologies Used

Technology	Purpose
MySQL	Database Management
SQL	Querying and Analysis
MySQL Workbench	Database Design & Execution

4. Database Name

CREATE DATABASE f1_championship;

Database Name: **f1_championship**

5. Database Schema

The project contains the following tables:

1. drivers
2. constructors

- 3. engines
- 4. seasons
- 5. driver_championship
- 6. constructor_championship

6. Table Description

6.1 Drivers Table

Stores information about Formula One drivers.

Column Name	Data Type	Description
driver_id	INT	Primary Key
driver_name	VARCHAR(100)	Driver Name
nationality	VARCHAR(50)	Driver Nationality

6.2 Constructors Table

Stores constructor/team information.

Column Name	Data Type	Description
constructor_id	INT	Primary Key
constructor_name	VARCHAR(100)	Team Name
country	VARCHAR(50)	Country

6.3 Engines Table

Stores engine manufacturer information.

Column Name	Data Type	Description
engine_id	INT	Primary Key
engine_name	VARCHAR(100)	Engine Manufacturer

6.4 Seasons Table

Stores season details.

Column Name Data Type Description

season_id	INT	Primary Key
year	INT	Championship Year
races	INT	Number of Races

6.5 Driver Championship Table

Stores championship-winning driver statistics.

Column Name Data Type Description

championship_id	INT	Primary Key
season_id	INT	Foreign Key
driver_id	INT	Foreign Key
constructor_id	INT	Foreign Key
engine_id	INT	Foreign Key
wins	INT	Number of Wins
poles	INT	Pole Positions
podiums	INT	Podium Finishes
points	DECIMAL	Championship Points
age	INT	Driver Age

6.6 Constructor Championship Table

Stores constructor championship winners.

Column Name Data Type Description

cc_id	INT	Primary Key
season_id	INT	Foreign Key
constructor_id	INT	Foreign Key

7. Entity Relationship Overview

Relationship between tables:

- One season can have one driver champion.
- One driver can win multiple championships.
- One constructor can win multiple championships.
- One engine can power multiple championship-winning cars.

Main relationships:

- driver_championship.season_id → seasons.season_id
- driver_championship.driver_id → drivers.driver_id
- driver_championship.constructor_id → constructors.constructor_id
- driver_championship.engine_id → engines.engine_id
- constructor_championship.constructor_id → constructors.constructor_id

8. Important SQL Queries

8.1 Driver With Most Championships

```
SELECT d.driver_name, COUNT(*) AS titles
FROM driver_championship dc
JOIN drivers d ON dc.driver_id = d.driver_id
GROUP BY d.driver_name
ORDER BY titles DESC;
```

Purpose:

Finds drivers with the highest number of championship titles.

8.2 Constructors With Most Championships

```
SELECT c.constructor_name, COUNT(*) AS titles
FROM constructor_championship cc
JOIN constructors c
ON cc.constructor_id = c.constructor_id
GROUP BY c.constructor_name
ORDER BY titles DESC;
```

Purpose:

Identifies the most successful constructor teams.

8.3 Driver Win Percentage

```
SELECT
d.driver_name,
s.year,
dc.wins,
s.races,
```

```
ROUND((dc.wins/s.races)*100,2) AS win_percentage
```

```
FROM driver_championship dc
```

```
JOIN drivers d ON dc.driver_id=d.driver_id
```

```
JOIN seasons s ON dc.season_id=s.season_id
```

```
ORDER BY win_percentage DESC;
```

Formula Used:

$$\text{Win Percentage} = \frac{\text{wins}}{\text{races}} \times 100$$

Purpose:

Calculates how dominant a driver was during a season.

8.4 Average Age of Champions by Decade

```
SELECT
```

```
FLOOR(s.year/10)*10 AS decade,
```

```
AVG(dc.age) AS avg_age
```

```
FROM driver_championship dc
```

```
JOIN seasons s ON dc.season_id=s.season_id
```

```
GROUP BY decade
```

```
ORDER BY decade;
```

Purpose:

Analyzes age trends among championship-winning drivers.

8.5 Engine With Most Championships

```
SELECT e.engine_name, COUNT(*) titles
```

```
FROM driver_championship dc
```

```
JOIN engines e ON dc.engine_id=e.engine_id
```

```
GROUP BY e.engine_name
```

```
ORDER BY titles DESC;
```

Purpose:

Finds the most successful engine manufacturers.

8.6 Driver Dominance Score

```
SELECT
```

```
d.driver_name,
```

```
s.year,
```

```
(dc.wins + dc.poles + dc.podiums) AS dominance_score
```

```
FROM driver_championship dc
```

```
JOIN drivers d ON dc.driver_id=d.driver_id
```

```
JOIN seasons s ON dc.season_id=s.season_id
```

```
ORDER BY dominance_score DESC;
```

Formula Used:

$$\text{Dominance Score} = \text{wins} + \text{poles} + \text{podiums}$$

Purpose:

Measures the overall dominance of a driver in a championship season.

9. Key Insights

- Michael Schumacher and Lewis Hamilton are among the most successful drivers in Formula One history.
 - Ferrari, McLaren, and Mercedes have dominated constructor championships.
 - Modern Formula One seasons have significantly more races compared to earlier decades.
 - Drivers in recent eras generally achieve higher point totals because of longer seasons and updated scoring systems.
 - Engine manufacturers such as Ferrari and Mercedes have had major historical success.
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10. Advantages of the Project

- Demonstrates practical use of SQL in sports analytics.
 - Uses relational database concepts effectively.
 - Performs meaningful statistical analysis.
 - Easy to extend with race-level data.
 - Useful for dashboard creation in Tableau or Power BI.
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11. Future Enhancements

Future improvements can include:

- Adding race-level statistics.
 - Including qualifying and lap-time data.
 - Creating stored procedures and views.
 - Integrating Tableau dashboards.
 - Building a web-based analytics system.
 - Using Python for predictive analytics and machine learning.
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12. Conclusion

The Formula One Championship Analysis project successfully demonstrates the use of SQL for managing and analysing historical sports data. The project provides insights into driver dominance, constructor performance, and championship trends across multiple decades.

It also highlights the importance of relational databases, foreign key relationships, joins, aggregation functions, and analytical SQL queries in real-world data analysis applications.